Executive Summary

Title:
TikTok
Project: Predicting Claim vs. Opinion

Subtitle

Data Examination

Project Overview

Through the initial EDA, an inspection and investigation of the data was conducted to better understand how the variables are correlated and the potential for unusually skewed output. The analysis provided an insight into the TikTok video features, the 298 null values that need to be discussed with the stakeholders, and uncovered **refinements and considerations for building a successful predictive model. Overall, the given dataset could effectively be used to train the model** to identify **claims or opinions, though room for improvement is always possible as more data is collected.**

Key insights

-The dataset is perfectly balanced, roughly 50/50 split between claims and opinions

- **The average view and share count of** videos **with "claim" status is much higher than videos with opinion status.**

- **The average view, like,** and **share counts of claim videos posted by banned accounts** are **higher than those posted by active and under review accounts.**

- **The likes per view and shares per view** are **also higher for claim** videos **posted by banned accounts than for active and under review accounts.**

- **The findings above indicate that the model should be** a tree-based**, either random forest model or XGBoost model, as they have the capabilities to handle outliers better than linear and logistic regression models.**

Details:

-Methodology: Import the necessary libraries, explore the data, inspect the descriptive statistics, check class balance, and examine the engagement trends associated with each claim status.

Variables Test:

Dependent variable: claim_status

Independent variables: video_duration_sec, video_view_count, video_share_count, verified_status, video_dowmload_count, video_comment_count, and author_ban_status.

Next Steps:

- **Further data cleaning and exploration need to be done.**
- **Drop features that do not provide value to the analysis or lead to an overfitted model.**
- **Train random forest trees and XGBoost and choose the one that would detect the most claims.**